IMMANUEL SOUNTHONEVICHIT

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SUMMARY

Mechanical Engineering with Robotics student (on track for a 1st) seeking a placement year in robotics, embedded or hardware engineering. Hands-on with embedded hardware, sensor integration and programming in C and Python. Runs Linux systems at the kernel and configuration level (Gentoo, Arch, Guix). Learns new tools quickly, works well in teams and takes ownership of problems.

EXPERIENCE

Manufacturing Workshop | University of Sussex, Brighton, UK | Nov 2025
- Machined a novelty hammer to specification on lathes and mills, following workshop safety procedures.

CAD Modelling | University of Sussex, Brighton, UK | Feb 2025
- Modelled a record player in SolidWorks and produced dimensioned technical drawings.
- Worked with a team of 4, splitting the workload efficiently.

Private Tutoring | Independent, London, UK | 2023
- Taught maths and English alongside A Levels, adapting explanations to different levels of understanding while managing the work around my own studies.

PROJECTS

Linux Systems | *Gentoo, OpenRC, Guix* | Ongoing
- Run Gentoo and Arch on a desktop and a ThinkPad X230, configuring hardened kernels, Portage, sysctl tuning, and Firejail/AppArmor sandboxing.
- Troubleshoot driver and kernel configuration issues. Currently learning Guile to build a GNU Guix System install.

CanSat - Main Engineer | *Python, Sensors, Teamwork* | Nov 2024 - Feb 2025
- Led integration of all sensors and the telemetry programming for a can-sized satellite, delivered to deadline alongside A Levels.
- Logged air pressure, temperature and humidity data in Python for analysis.

Self-Hosted Homelab | *Alpine Linux, WireGuard* | Ongoing
- Building a homelab on a Raspberry Pi 4 running Alpine Linux, with WireGuard for secure remote access.
- Configured a router and port forwarding to expose only required services.

HackSussex Hackathon | *Godot, Teamwork* | Feb 2026
- Made a rhythm game in 24 hours in a team of 3, splitting tasks and time-boxing to produce a working demo.
- Debugged collaboratively through pair programming.

Global Design Challenge | *Teamwork* | Jan 2026
- Worked in a team of 5 to design an app for circulating second-hand items, balancing sustainability, user needs, and practical constraints.
- Pitched the concept to a panel, communicating the rationale clearly and responding to questions.

SKILLS

- **Embedded & Hardware:** Sensor integration, Raspberry Pi, basic electronics, hardware troubleshooting
- **Software & Systems:** C, Python, C#, Shell, Git, Linux + GNU (Gentoo, Arch, Alpine, Guix), Godot, WireGuard
- **Data & Analysis:** MATLAB, Excel, data logging, data analysis
- **Engineering:** SolidWorks, technical drawing, prototyping, lathe and milling, workshop safety

EDUCATION

University of Sussex | Brighton, UK | Sep 2025 - Present
BEng Mechanical Engineering with Robotics (IMechE accredited)
- On track for a 1st, 76% stage average
- **Relevant modules:** Programming for Engineers, Electrical Circuits & Devices, Engineering Maths, Engineering Mechanics, Technical Drawing & CAD, Materials & Manufacturing Processes, Thermodynamics, Fluid Mechanics

Wren Academy | London, UK | Sep 2023 - Jul 2025
A Levels: Computer Science, Physics, Maths